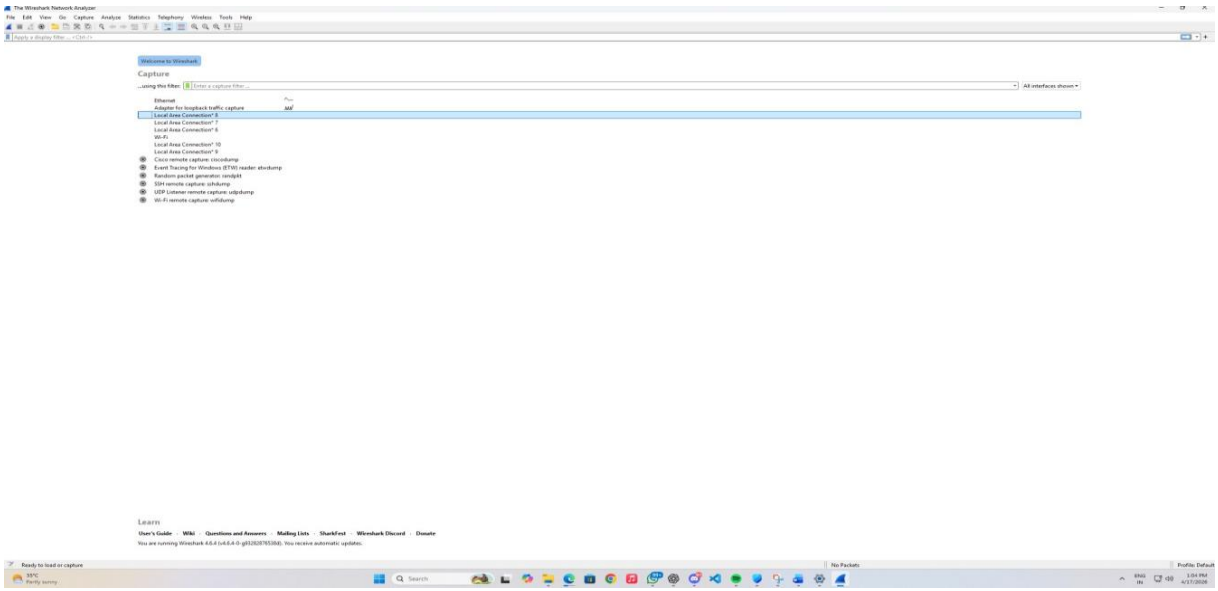
1. To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

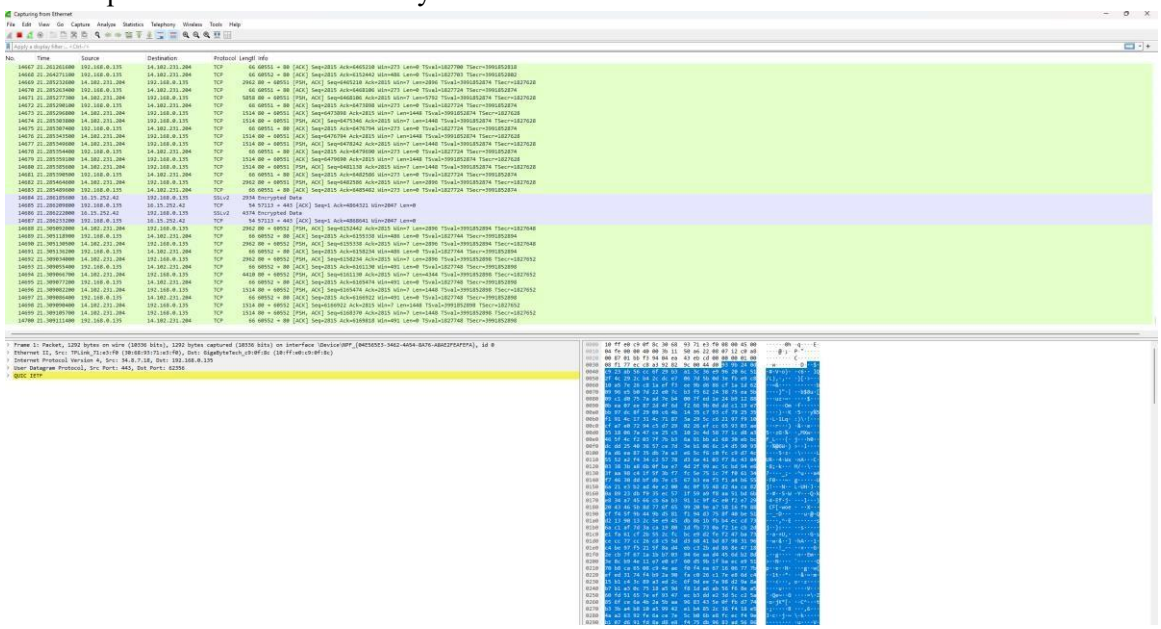
Step 1: Launch Wireshark

- Open **Wireshark**
- You will see a list of **network interfaces**
  - Wi-Fi
  - Ethernet



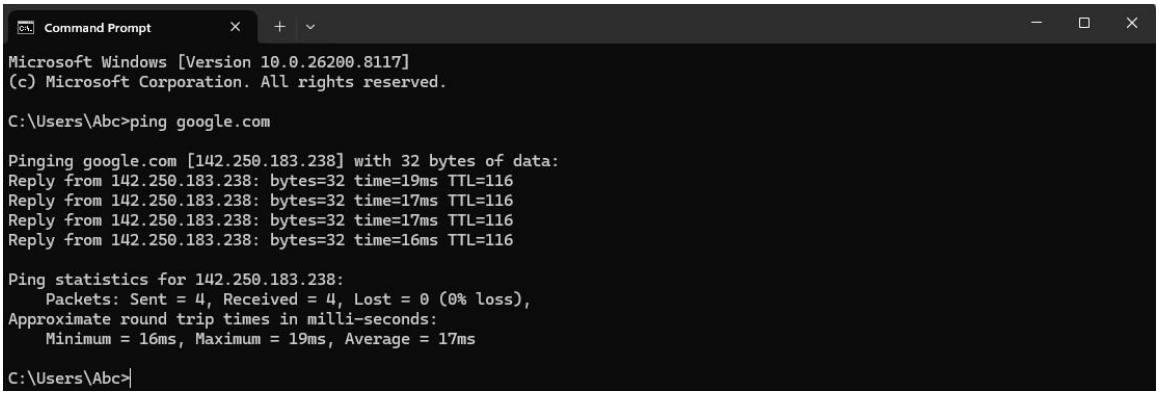
Step 2: Start Packet Capture

- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately



Step 3: Generate Network Traffic Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com) □ Open Command Prompt and run:
- ping google.com



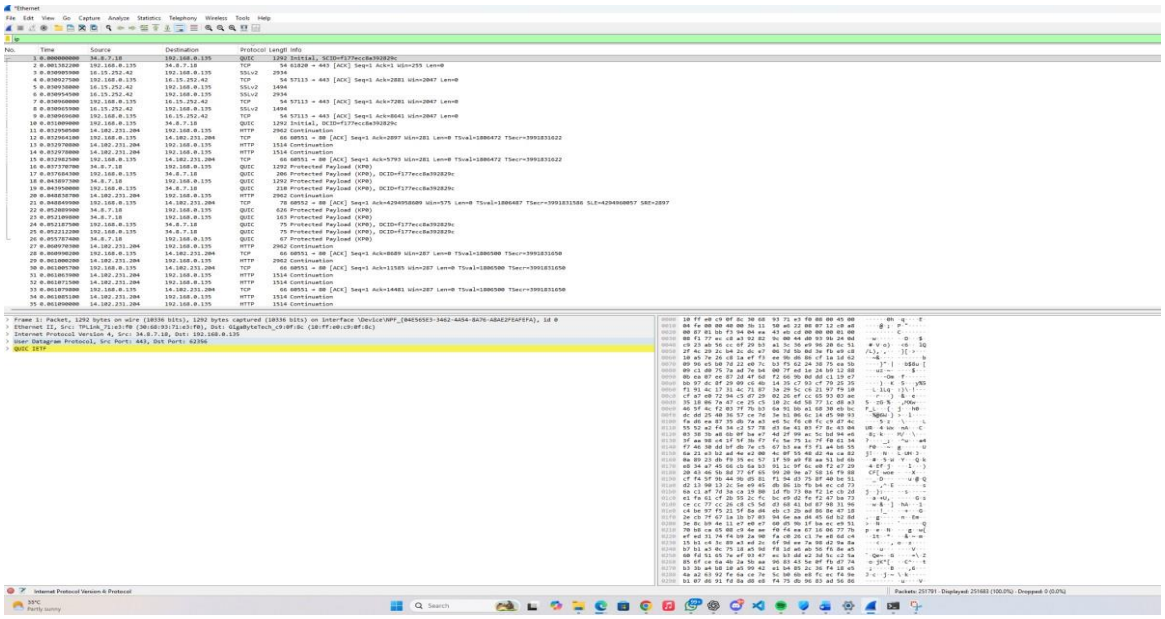
Step 4: Stop Capture

- Click the **red Stop button**
- Captured packets will remain on the screen

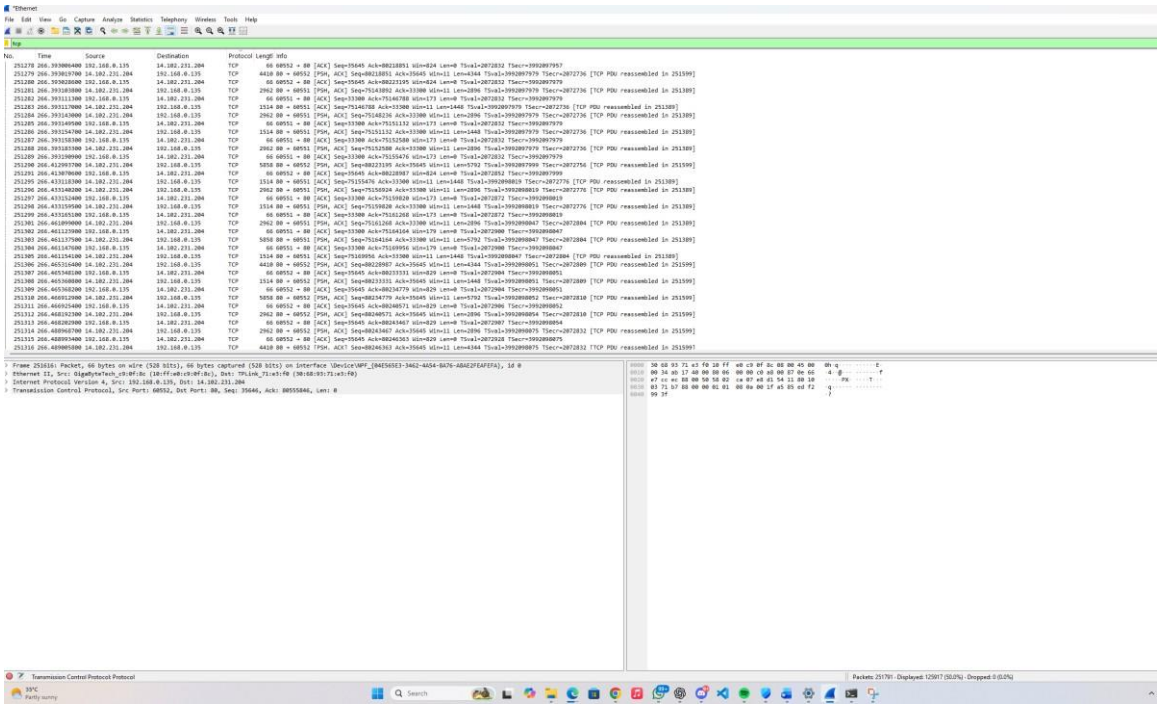
Step 5: Apply Basic Filters Use the **Display Filter bar**:

Filter Purpose ip

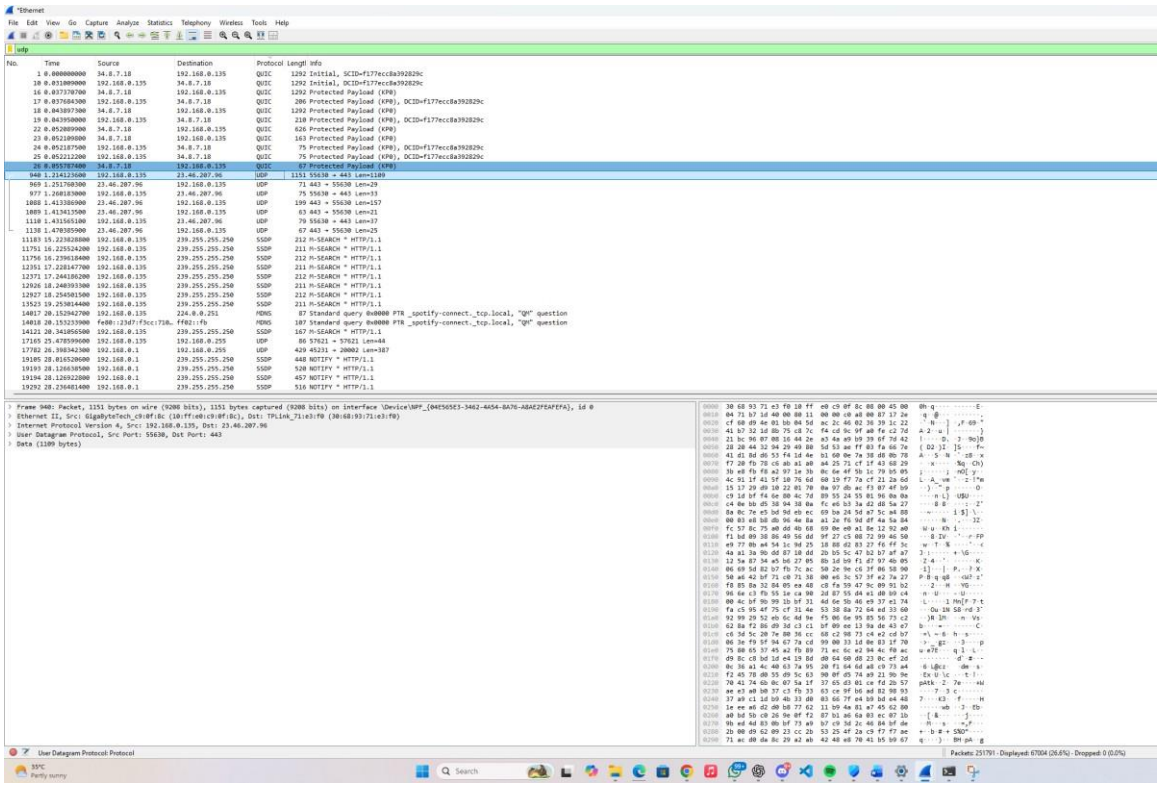
Show IP packets



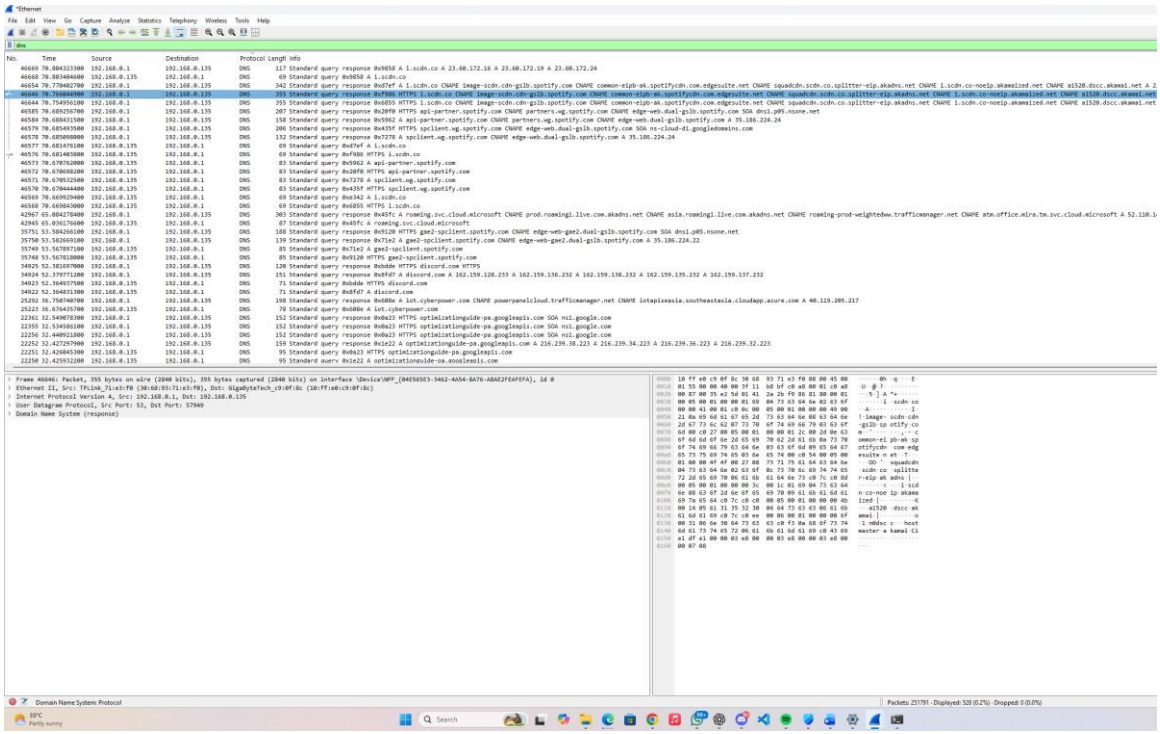
- tcp
- Show TCP packets



- udp
- Show UDP packets

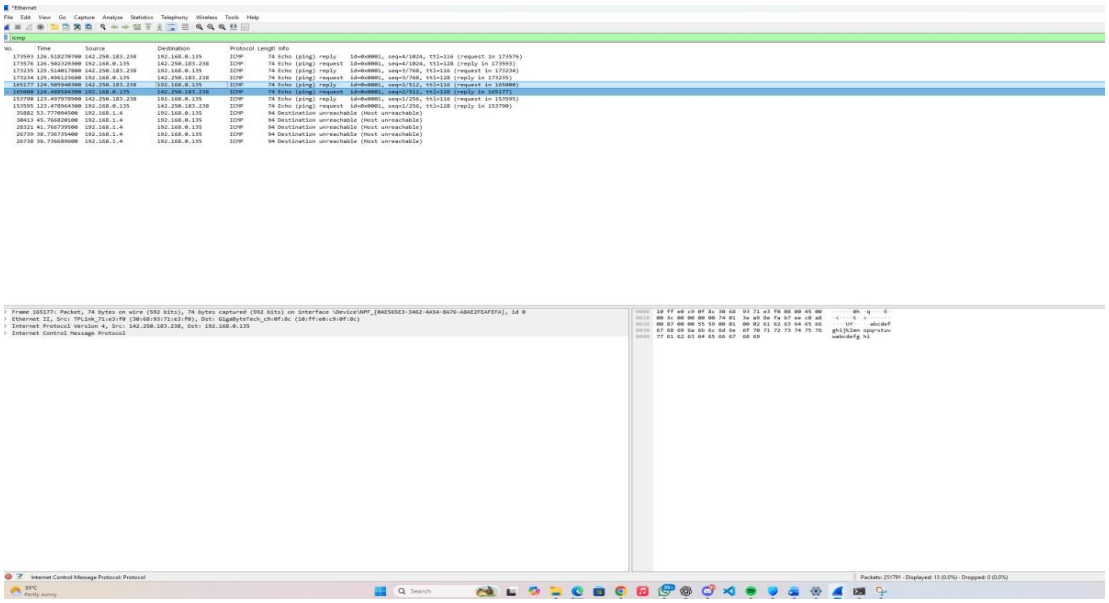


- dns
- Show DNS packets



icmp

Show ping packets



Step 6: Analyze a Packet

- Click on any packet
  Observe:

Ethernet header-

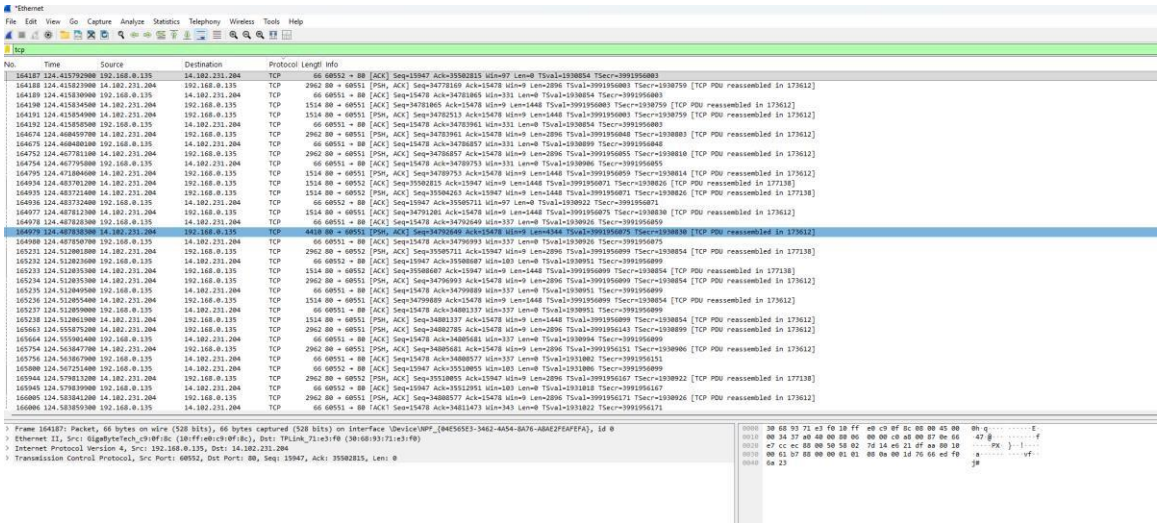
Source MAC Address: ec:e0:c9:0f:8c:08
Destination MAC Address: 30:68:93:71:e3:f0
Type:IPv4

IPheader-

Source IP Address: 192.168.0.135
Destination IP Address: 14.102.231.204
Protocol:

TCP/UDP header-

Source Port: 60552
Destination Port: 80 (HTTP)
Sequence Number: 15947
Acknowledgment Number: 35502815
Flags: ACK
Payload data-Length**:** 0 bytes (Len = 0 visible)



Observation

- Interface used: Ethernet (Local Area Connection)
- Protocols observed: TCP, UDP, QUIC, SSDP, MDNS, IP
- Source and destination IP addresses: 192.168.0.135, 14.102.231.204 Packet length: **:** 66 bytes

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.